

# Features: Useful Clues

Project: Helpful Classroom Sorter

## 1. Your mission

### Your learning target

I can choose recorded properties that preserve distinctions needed for a stated task.

I can show why identical feature records can hide examples with different target labels.

I can reject an answer copied into the inputs and test a proposed feature with a new case.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Input, Process, Output. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Some observable clues help a decision more than others.



LEARNING TARGET A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

Learner name:

## Readiness for the connected explanation

Two annotators agree that a calendar card is ready. What must you still check before treating that label as correct?

## This week's end goal

By the end, I can choose task-relevant features and avoid a clue created after the decision.

## 2. Notice the evidence

PICTURE STORY

### Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY Ear shape, muzzle length, and paw shape may help distinguish cats from dogs; background color and photo border usually do not.

### What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

### A closer explanation

Imagine a paper sorting task for a model delivery station. Each rectangular card has a length, width, colour and card number. The station labels a card fits if its length is at most 12 centimetres and its width is at most 8 centimetres. Both limits include equality. Cards stay flat in the marked orientation; turning, bending and thickness are outside this paper task. All measurements in the first examples are known positive whole numbers.

A feature is a recorded input property, such as width in centimetres. The target label is the answer we want to produce: fits or too large under this guide. Decide what the answer means before choosing the inputs. A card number can help us find a record, but knowing its number does not tell us its dimensions.

This guide already gives us a direct rule, so we do not need to train a model to solve it. We use the task to

investigate the information that either a rule or a learned model would need. Later, when relationships are learned from examples, the same question remains: did we supply the distinctions the task requires?

### 3. See how it works

VISUAL MECHANISM

**Input → Process → Output**

Follow the information instead of imagining magic inside the machine.



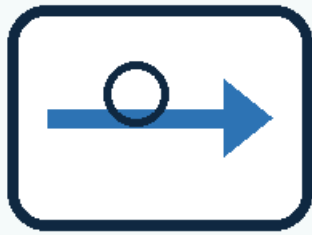
**INPUT**

measured candidate clues



**PROCESS**

compare useful patterns and ignore  
distractions



## OUTPUT

predicted cat/dog label with evidence

INPUT measured candidate clues

PROCESS compare useful patterns and ignore distractions

OUTPUT predicted cat/dog label with evidence

**CORE IDEA** Some observable clues help a decision more than others.

**UNDER THE HOOD** A feature is a represented property used by a decision process. Predictive usefulness depends on the task and data. Spurious features correlate in a sample without reflecting the intended concept and often fail when conditions change.

Draw a fresh example with all three stages:

### A closer explanation

Card A is 10 centimetres long and 7 centimetres wide. Card B is 10 centimetres long and 9 centimetres wide. A fits; B is too large because its width exceeds 8. Compare two ways of recording these cards.

### The same cards under two feature choices

**Original card · Choose the inputs · Resulting feature record**

**Original card:** A is 10 by 7 cm and fits

**Choose the inputs:** Keep length only

**Resulting feature record:** 10 cm

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**Original card:** B is 10 by 9 cm and is too large

**Choose the inputs:** Keep length only

**Resulting feature record:** 10 cm

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**Original card:** A is 10 by 7 cm and fits

**Choose the inputs:** Keep length and width

**Resulting feature record:** 10 cm and 7 cm

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**Original card:** B is 10 by 9 cm and is too large

**Choose the inputs:** Keep length and width

**Resulting feature record:** 10 cm and 9 cm

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Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

A system receiving only the length cannot use the hidden width to tell these two cards apart. The same input, 10, is paired with two different correct answers. It may guess, return an uncertainty, or use a usual answer, but length alone cannot guarantee the right label for both. A more complicated model cannot reconstruct this missing distinction with certainty from that input alone.

A feature record needs a consistent order and units. Here we write length in centimetres first, then width in centimetres. The record 10, 7 means something different if someone silently switches the order or records millimetres. Write the definitions alongside the data.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

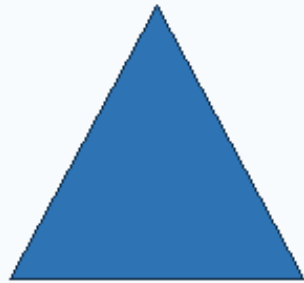
PICTURE GALLERY

### Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



**Ear shape**



**Muzzle length**



**Photo border**



**Grass background**

CASE 1 — Ear shape Clue: may differ across many examples. Meaning: potentially useful.

CASE 2 — Muzzle length Clue: observable animal trait. Meaning: potentially useful.

CASE 3 — Photo border Clue: added by the photographer. Meaning: distraction.

CASE 4 — Grass background Clue: accidental setting pattern. Meaning: spurious clue.

**BOUNDARY** A clue can be observable yet still be irrelevant, unfair, or fragile.

Which two pictures are easiest to confuse? What evidence separates them?



## Change one thing in the pictured case

Change: Change Q's width to 6 without changing its height.

Prediction: Q fits the shelf.

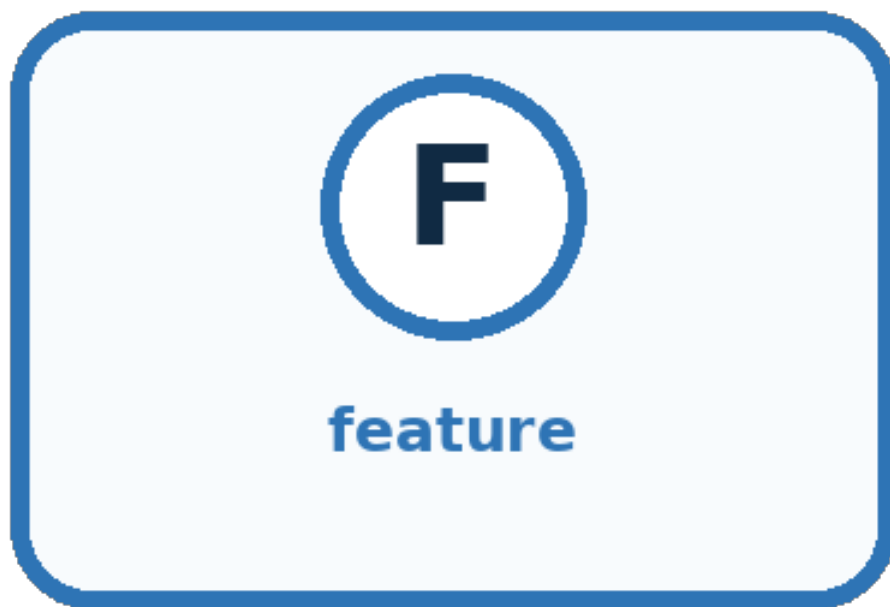
Reason: Both dimensions now meet their including equality limits; stripe pattern does not change the rule.

## 5. Words that unlock the idea

### VOCABULARY

#### Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





**relevant**



**distraction**



### Word · Meaning · My example

**Word:** feature

**Meaning:** represented clue used in a decision

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Word:** relevant

**Meaning:** connected to the task goal

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Word:** distraction

**Meaning:** available clue that should not drive the decision

**My example:** \_\_\_\_\_

\_\_\_\_\_

**Word:** spurious clue

**Meaning:** accidental pattern that may fail later

**My example:** \_\_\_\_\_

\_\_\_\_\_

### Misconception detective

**CLAIM TO REPAIR** Use every available clue because more information is always better.

Use one vocabulary word and one piece of evidence to repair it:

**WORD POWER** A strong definition helps you decide something in a new case.

## Words used in the closer explanation

### Word · Meaning

**Word:** Target label

**Meaning:** The answer a supervised learning task asks a model to predict

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**Word:** Feature record

**Meaning:** The selected input values for one example, in a defined order

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**Word:** Relevant feature

**Meaning:** An input property that helps make distinctions needed for the stated task

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**Word:** Leakage

**Meaning:** Information used during development that would not legitimately be available when making the intended prediction

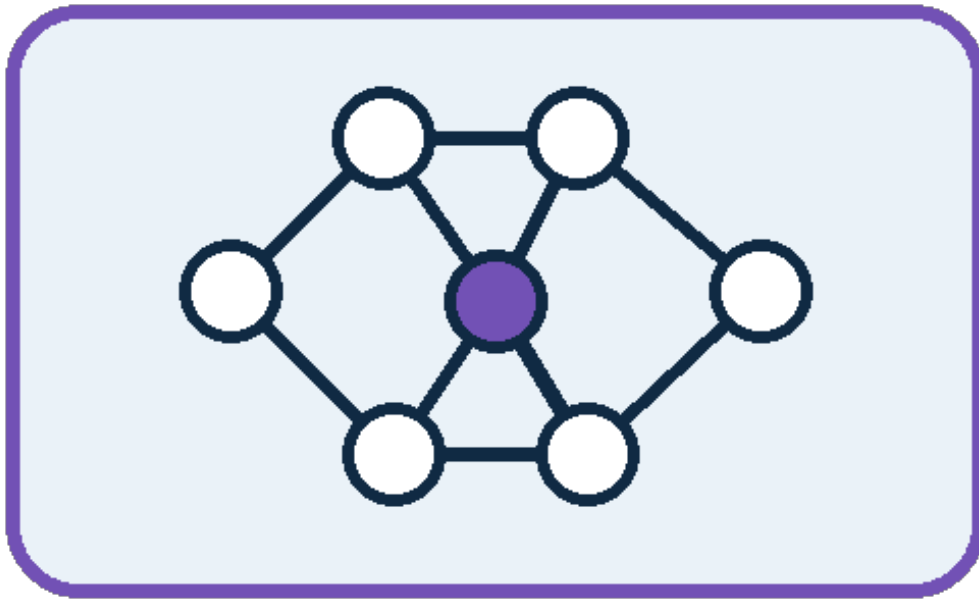
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## 6. Follow a worked example

WORKED EXAMPLE

### Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE Ear shape, muzzle length, and paw shape may help distinguish cats from dogs; background color and photo border usually do not.

STEP 1 — OBSERVE measured candidate clues

STEP 2 — TRACE compare useful patterns and ignore distractions

STEP 3 — CONCLUDE predicted cat/dog label with evidence

MODEL REASONING Useful features describe the animal and remain available across settings. Borders and backgrounds describe the photo setup. Even ear shape may fail for unusual breeds or blocked views, so multiple clues and uncertainty are needed.

DO NOT OVERCLAIM A clue can be observable yet still be irrelevant, unfair, or fragile.

What new evidence could change this answer?

### A closer explanation

**Card measurements and colour · Label under the guide · What the comparison shows**

**Card measurements and colour:** 10 by 7 cm and blue

**Label under the guide:** Fits

**What the comparison shows:** Compare with the next card

**Card measurements and colour:** 10 by 9 cm and blue

**Label under the guide:** Too large

**What the comparison shows:** Same length and colour but width changes the label

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**Card measurements and colour:** 13 by 7 cm and blue

**Label under the guide:** Too large

**What the comparison shows:** Same width and colour as the first card but length changes the label

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**Card measurements and colour:** 10 by 7 cm and red

**Label under the guide:** Fits

**What the comparison shows:** Changing only colour does not change this guide

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The first two cards show why width matters. The first and third show why length matters. The first and fourth show that colour has no part in this particular rule. That is a statement about our task, not a claim that colour is useless for every task. For sorting red and blue cards, colour would be essential.

Suppose every fitting card in a small practice set is blue and every oversized card is red. Colour would appear to predict the labels perfectly in that set. A red 10 by 7 card and a blue 13 by 7 card expose the shortcut. Check a proposed feature on deliberately different cases before trusting the practice score.

## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Features: Useful Clues

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

### The facts we will use

Boxes  $P=(6,4,\text{stripes})$ ,  $Q=(7,4,\text{stripes})$ ,  $R=(6,5,\text{dots})$ ,  $S=(6,4,\text{dots})$ .

A green acceptance sticker is attached only after the shelf check.

## Dimensions determine shelf acceptance

|           | Width cm | Height cm | Outcome      |
|-----------|----------|-----------|--------------|
| P stripes | 6        | 4         | fits         |
| Q stripes | 7        | 4         | width fails  |
| R dots    | 6        | 5         | height fails |
| S dots    | 6        | 4         | fits         |

Read a cell using both its row and column labels.

## Features: Useful Clues

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Width check

P,Q,R,S have widths 6,7,6,6.



#### 2. Height check

Their heights are 4,4,5,4.



#### 3. Apply both limits

P and S fit; Q fails width; R fails height.



#### 4. Reject a shortcut

The later sticker cannot be an input available before checking.

### CHANGE ONE THING

Change Q's width to 6 without changing its height.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Width check

**What happens and why:** P,Q,R,S have widths 6,7,6,6.

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**Step:** 2. Height check

**What happens and why:** Their heights are 4,4,5,4.

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**Step:** 3. Apply both limits

**What happens and why:** P and S fit; Q fails width; R fails height.

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**Step:** 4. Reject a shortcut

**What happens and why:** The later sticker cannot be an input available before checking.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Comparing P with Q isolates width; comparing P with R isolates height. Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding feature. A later acceptance sticker leaks the very answer the system should produce.

### What this example does not establish

These comparisons establish the written shelf rule, not the importance of pattern for unrelated tasks.

## 7. Find and repair a mistake

COMPARE

### Same Result, Different Reason

A comparison helps reveal the true concept boundary.

#### Case · Evidence · What it shows

**Case:** Ear shape

**Evidence:** may differ across many examples

**What it shows:** potentially useful

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**Case:** Muzzle length

**Evidence:** observable animal trait

**What it shows:** potentially useful

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**Case:** Photo border

**Evidence:** added by the photographer

**What it shows:** distraction

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**Case:** Grass background

**Evidence:** accidental setting pattern

**What it shows:** spurious clue

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## Repair two tempting claims

CLAIM A Use every available clue because more information is always better.

Repair:

CLAIM B A feature that worked once will always work.

Repair:

STRONG ANSWER FRAME I claim \_\_\_\_\_ because the evidence shows \_\_\_\_\_. The process is \_\_\_\_\_. One limit is \_\_\_\_\_.

## A closer explanation

Before adopting a feature set, state the task, name the inputs with units, and ask whether two different answers can share the same input record. Construct such a pair if you can. Then check that each input is available at the time the answer is needed. These checks can reveal a broken representation before any model is trained.

For our known measurements and fixed orientation, length and width are sufficient to apply the paper rule exactly. That does not establish whether a real parcel will fit through a real opening: thickness, shape and measurement error were excluded. If the task changes, review the features again. More columns are not automatically better; each one needs a reason to be collected and used.

**Someone claims that a model using only card colour is reliable because it matches every practice label. Give a new card that challenges this claim and explain what should be checked instead**

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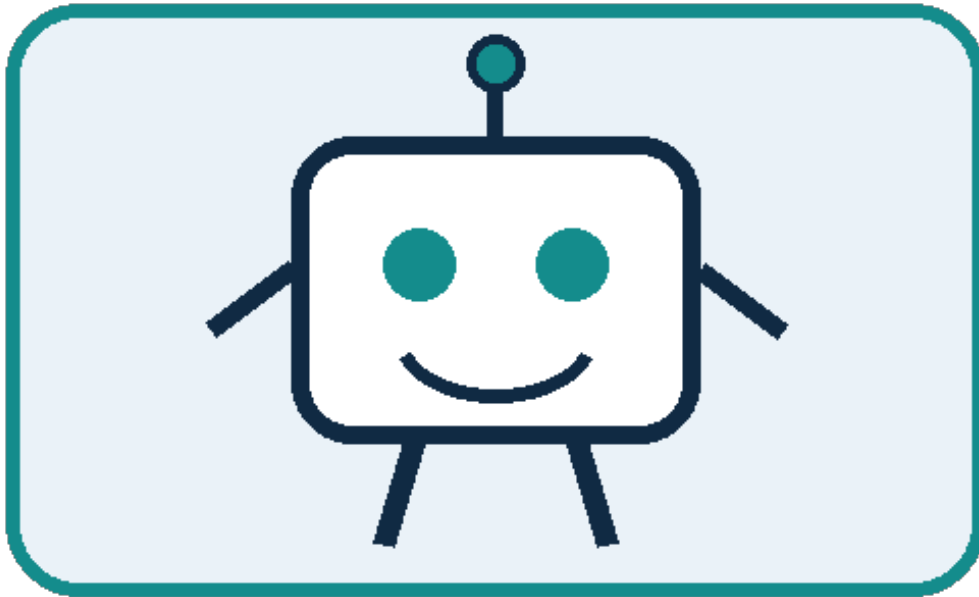
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## 8. Practise together

## GUIDED PRACTICE

### Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Some observable clues help a decision more than others.

Use a fresh example to explain feature.

Trace a new input → process → output example.

Find and repair the error in: A feature that worked once will always work.

Invent one edge case and name the safe response.

**PARTNER CHECK** Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

### A closer explanation

After measuring a card, a teacher adds a sticker saying fits or too large. A learner proposes using sticker text as an input to predict the label before that inspection. The sticker seems excellent in the completed records because it contains the answer. But it does not exist at the time of the intended prediction. This is leakage: development has used information the real task cannot provide.

Draw a line at the moment the system must answer. On the before side, put the permitted measurements available then. On the after side, put the final label sticker. Keep labels available for teaching and checking predictions, but do not feed the sticker back in as an input for this task.

**A card is 12 cm long and its width is unreadable. Someone proposes using the label sticker that will be added after inspection. Which information is usable before inspection? Explain why it does not establish the label**

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### Practise with less help: Prove that one measurement is insufficient

Use workbook W2 for the connected example “Features and useful distinctions”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Features and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Point to the pair that tests pattern, then explain why the sticker is a bad early input.

#### My explanation using a pictured step and the given facts

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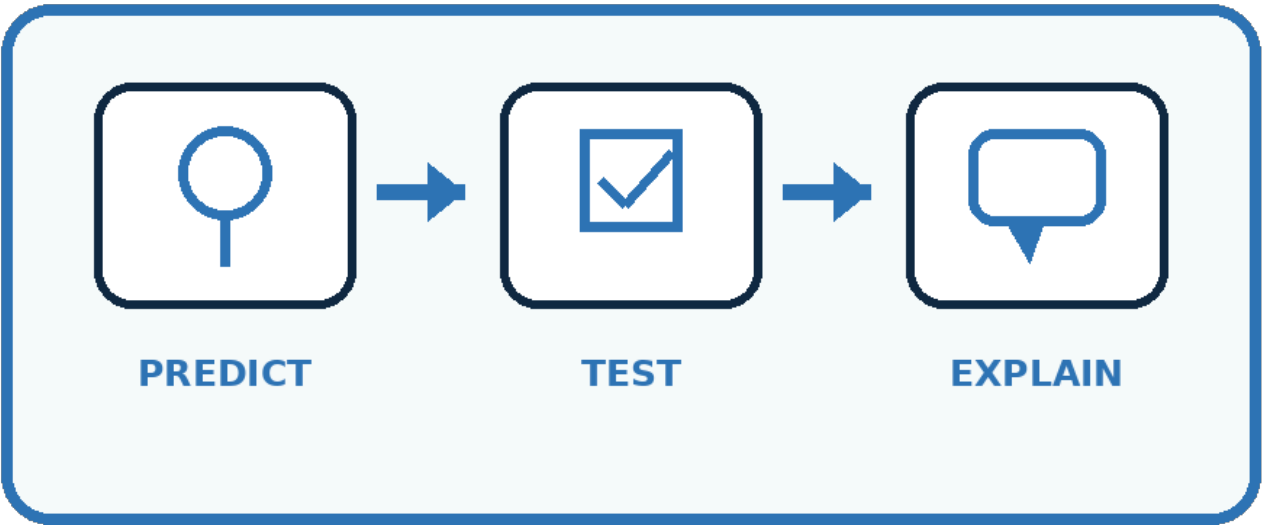
This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

EXPLORER LAB

### Clue Keep-or-Reject

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Choose three clues for cat/dog sorting; reject distractions.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

**Case · Prediction · Observed result · What it means**

**Case:** Normal

**Prediction:** \_\_\_\_\_

**Observed result:** \_\_\_\_\_

**What it means:** \_\_\_\_\_

Case: Edge

Prediction: \_\_\_\_\_

Observed result: \_\_\_\_\_

What it means: \_\_\_\_\_

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

### Plan → try → check

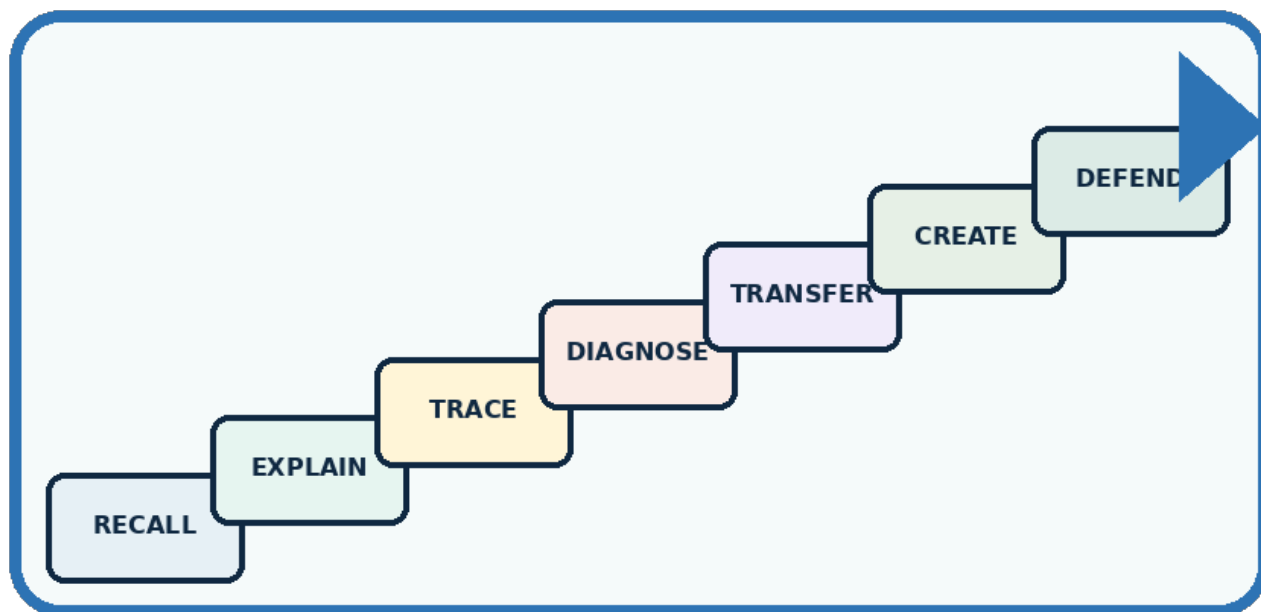
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

PRACTICE LADDER

### Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define feature.

EXPLAIN Some observable clues help a decision more than others.

TRACE Show the input, process, and output.

**DIAGNOSE** Repair: Use every available clue because more information is always better.

**TRANSFER** Apply the idea to a new home or classroom case.

**CREATE** Design one revealing test.

**DEFEND** Give a claim, evidence, mechanism, limit, and safe action.

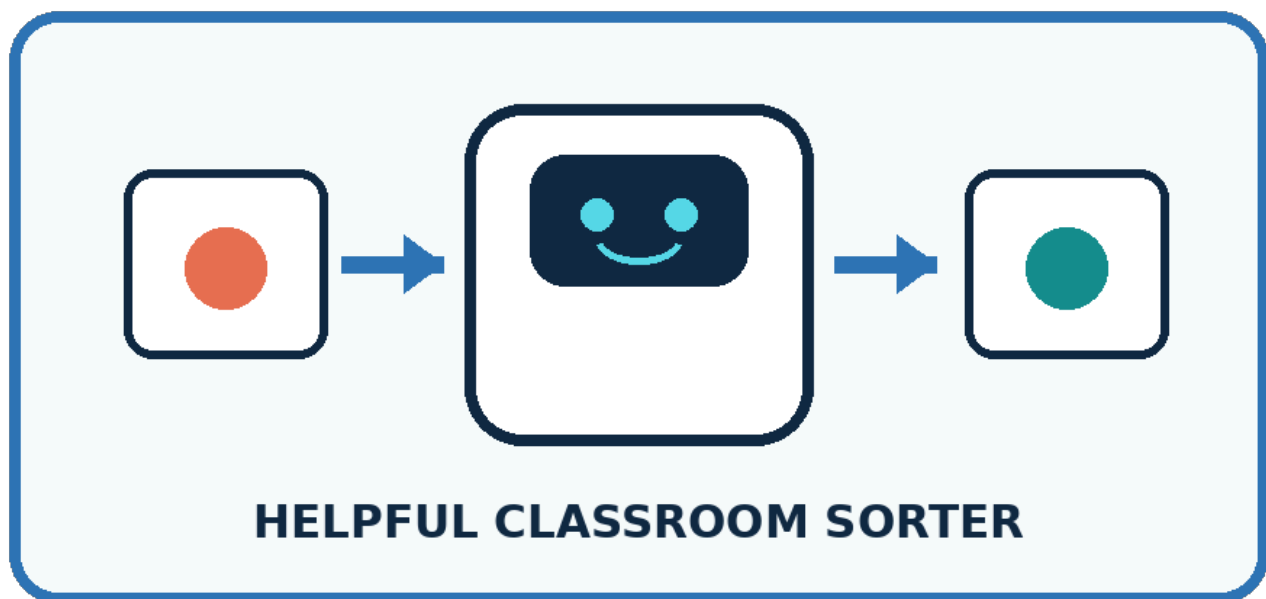
**RETRY RULE** A coached answer becomes mastery only after you solve a different case independently.

## 11. Connect your project

**PROJECT**

### Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



**THIS WEEK'S CHECKPOINT** Select the smallest useful features for lost-and-found sorting and reject clues tied to a particular classroom or owner.

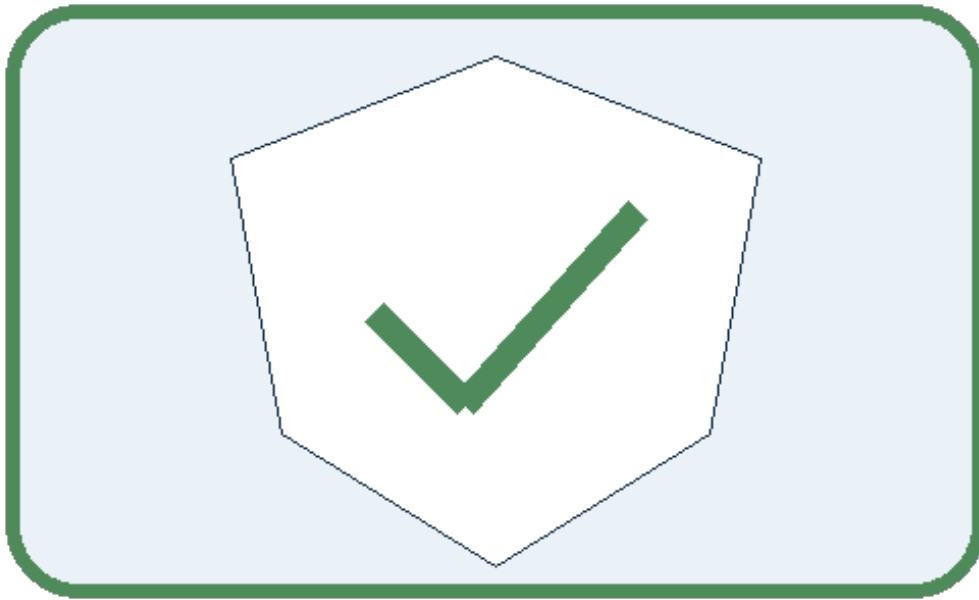
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN A clue can be observable yet still be irrelevant, unfair, or fragile.

My project decision:

### **Evidence for your project**

Choose sorter features that exist at decision time and actually connect to the stated classroom requirement.

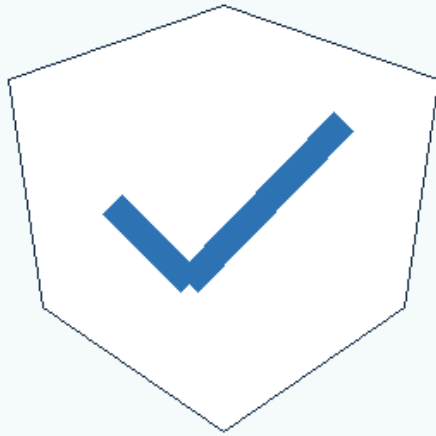
## **12. Show what you know**

MASTERY

### **Fresh-Case Independent Proof**

Complete this after teaching and guided practice have ended.





## INDEPENDENT PROOF

**REQUIRED PROOF** Select three features for new cat/dog cards, reject two distractions, and predict one case where a chosen feature would fail.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

**SELF-CHECK** Useful features describe the animal and remain available across settings. Borders and backgrounds describe the photo setup. Even ear shape may fail for unusual breeds or blocked views, so multiple clues and uncertainty are needed.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

**NEXT BRIDGE** Carry today's evidence habit into: Algorithms as Recipes.

## 13. Study a complete case

### Week 8 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This

case extends the illustrated lesson.

**The situation**

Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

**The question**

Choose the relevant feature and explain why size is insufficient.

**Worked explanation**

Edge shape directly addresses the target. Size can vary within both categories. Feature usefulness depends on the task.

**Explain it in your own words**

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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**Trace the supplied case visually**

**Choose a clue that answers this task**

**Evidence or stage · Value or result**

**Evidence or stage:** Target

**Value or result:** Smooth versus toothed edge

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**Evidence or stage:** Useful clue

**Value or result:** Edge shape

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**Evidence or stage:** Weak clue

**Value or result:** Leaf size alone

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Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

Choose the relevant feature and explain why size is insufficient.

Edge shape directly addresses the target. Size can vary within both categories. Feature usefulness depends on the task.

# 14. Try a new case independently

## Independent case: Design a record for a new task

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name. A pass sticker is added afterward. Choose inputs with units. Construct two tiles with the same height but different labels. Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an input before checking.

### My independent explanation, trace and limits

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Week 8 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A sorter separates round and square tiles. Both shapes come in red and blue. Choose a useful feature and a distracting feature. Test your choice on a blue circle.

My answer, with a trace, calculation or supporting evidence

### Write or draw your reasoning and evidence.

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**Create and test**

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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**Connect to your project**

Select the smallest useful features for lost-and-found sorting and reject clues tied to a particular classroom or owner.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

**Transfer practice from the original programme**

A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

**Explain the mechanism, evidence and one limit.**

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### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.